

AI Assistant Used: Claude (Anthropic)

Problem 7, Task 1: Rotation Sandbox (Current vs. Fixed Frame)

Prompt 1 — Core Composition Logic:

"I understand that composing rotations about the current (body) frame uses right-multiplication ($R_{\text{new}} = R_{\text{old}} @ R_{\text{step}}$) and composing about the fixed (space) frame uses left-multiplication ($R_{\text{new}} = R_{\text{step}} @ R_{\text{old}}$). Can you write a Python function that takes a sequence of (axis, angle, frame) tuples and applies these rules in order, returning the final rotation matrix?"

Claude Output:

Provided a `compose_sequence()` function with correct if/elif logic for the two multiplication rules.

What I Verified & Changed:

Tested the function against my Problem 3 symbolic solution (two 90° rotations about Z then X). Confirmed the final matrices matched my hand-calculated results ✓ No changes needed; the logic was correct.

Prompt 2 — Interactive UI with Sliders and Frame Toggle:

"I want a Tkinter UI with three sliders (one per axis, range -180 to 180 degrees) and radio buttons per axis to toggle "Current" vs "Fixed" frame. The MuJoCo viewer should update in real time as I move sliders. The current orientation and frame settings should display as a readable sequence."

Claude Output:

Provided `03_rotation_ui.py` with a `RotationState` class, slider callbacks, and live matrix printout.

What I Verified & Changed:

Ran the script and confirmed that sliders update the viewer in real time. Toggled frame buttons and visually confirmed the orientation changed. Added order-slot dropdowns (user-requested feature) to show that rotation order also matters. Verified numeric labels match slider positions

Prompt 3 — Auto-Play Animation for Recording:

"Now add two buttons, "Play Sequence 1 (current frame)" and "Play Sequence 2 (fixed frame)", that automatically animate the sliders over ~1.2 seconds per axis. After one axis finishes, move to the next. This should reset everything to 0 first, then smoothly move each slider to its target angle."

Claude Output:

Rewrote the UI to include the `play_sequence()` method, which schedules animations recursively.

What I Verified & Changed:

Tested both buttons; animations ran smoothly and reset correctly. Edited `SEQUENCE_1` and `SEQUENCE_2` at the top of the file to match my exact Problem 3 example (Z 90° , X 90°). Recorded one continuous video showing both sequences playing back-to-back

Problem 8, Task 2: Verify Skew-Symmetric Identities in Simulation

Prompt 1 — Setting Up MuJoCo Simulation with Angular Velocity:

"I want to load a MuJoCo model, set the body to spin with constant angular velocity $\omega = [0.1, 0.05, 0.15]$ rad/s, step the simulation, and log the body's rotation matrix R at several time points ($t = 1, 2, 3, 4, 5$ seconds). Show me how to access R from `mujoco.MjData` and when to call `mujoco.mj_step()`."

Claude Output:

Provided simulation loop with `mj_step()`, `data.body("body_name").xmat` extraction, and time-triggered logging.

What I Verified & Changed:

Ran the simulation and confirmed that R was orthogonal ($R @ R.T \approx I$) at each logged step. Body rotated as expected. Used the `asymmetric_body.xml` model from the starter code. No changes needed

Prompt 2 — Implementing Identity Checks:

"For each logged R matrix, I want to: 1. Draw 5 random vectors v, w, ω from a normal distribution 2. Compute both sides of $R(v \times w) = (Rv) \times (Rw)$ and measure residual $||LHS - RHS||$ 3. Compute both sides of $R \hat{\omega} R^T = (R\omega)^\wedge$ using max-norm on elements 4. Track the worst-case residual across all 5 trials per time step. Can you provide the test functions?"

Claude Output:

Provided `check_cross_product_identity()` and `check_skew_conjugation_identity()` with random sampling and residual computation.

What I Verified & Changed:

Tested with my own hand-crafted test vectors ($v=[1,0,0]$, $w=[0,1,0]$) and confirmed logic. Ran the full suite 5 times per time step as specified. Cross-checked residual formulas against the Problem 5 identity statements

Prompt 3 — Analysis and Report Writing:

"All my residuals came out at $\sim 1e-16$ (machine epsilon). Help me interpret why this residual doesn't prove the identity? What does it mean if residuals were consistently $1e-8$ instead? Write a brief interpretation."

Claude Output:

Provided interpretation distinguishing numerical evidence from mathematical proof, explaining floating-point precision limits, and the implications of larger residuals.

What I Verified & Changed:

Read the interpretation carefully and re-verified my understanding of machine epsilon. Added my own explanation of why finite sampling doesn't constitute proof. Created final report document (`Q8_Report.docx`) with tables and conclusions

Problem 9, Task 3: Current vs. Fixed Frame in ROS/RViz

Prompt 1 — ROS 2 Node with Parameter Support:

"I have a ROS 2 starter template that declares a parameter `compose_frame` with values "current" or "fixed". Show me how to read this parameter inside the `on_timer` callback and use it to choose between $R_{body} = R_{body} @ R_{step}$ (current) and $R_{body} = R_{step} @ R_{body}$ (fixed)."

Claude Output:

Provided the if/elif logic in `on_timer()` exactly mirroring the MuJoCo solution from Problem 7.

What I Verified & Changed:

Built the workspace: `colcon build --symlink-install`. Ran the node and tested parameter toggling live. Confirmed that changing `compose_frame` updated the `body_frame` TF transform immediately. Code was already correct; no changes needed

Prompt 2 — Quaternion Conversion for TF:

"The starter code provides `R_to_quat_xyzw()` to convert rotation matrices to quaternions in ROS XYZW order. Is this correct, and why is the order different from MuJoCo (which uses WXYZ)?"

Claude Output:

Confirmed the conversion is correct per ROS standards, explained the historical reason for the difference, and noted it in the README.

What I Verified & Changed:

Didn't need to change it; just noted the difference for my own documentation ✓ Tested the node output against RViz2 and confirmed transforms looked correct

Prompt 3 — Recording and Demo Script:

"Walk me through the exact terminal commands to: 1. Start the `tf_broadcaster_node` 2. Open RViz2 and set up the TF display 3. Toggle the `compose_frame` parameter to show both cases 4. Know when to start/stop a screen recording to capture both"

Claude Output:

Provided step-by-step commands for three terminals, timing guidance, and rviz setup instructions.

What I Verified & Changed:

Executed the exact commands in order. Recorded a video showing `space_frame` (static) and `body_frame` (rotating) under both modes. Wrote accompanying 2-3 paragraph explanation tying the visual demo to the math

Final Prompt: Prepare a report for me of all the conversations we had between you and me in the format prompt. What was your output for each question that we sequentially followed?

Claude Output: all the text of the prompt and what Claude output is.

What I Verified and Changed:

I verified each prompt first, then read the report and added a write-up of what I verified after receiving the output.

